LECTURE 09 · REVIEW NOTES

Equity valuation: ratios & technical analysis

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In our previous lectures, we valued equity using dividends and FCFE. We started with the cash shareholders could receive and asked what that cash was worth today. Here, I want you to approach the same valuation question from a different direction: how much does the market pay for a similar business?

Think of a multiple as a price tag attached to one dollar of a financial measure. A P/E of 30× means investors pay $30 for each $1 of annual earnings. It does not mean investors will recover their money in exactly 30 years. Earnings can change, and earnings are not necessarily distributed to shareholders.

1. First, be clear about the terms

EPS means earnings per share. It allocates net earnings attributable to common shareholders across weighted-average shares. Diluted EPS also reflects potentially dilutive securities. **Market capitalization** is the market value of all outstanding shares. A price of $200 is the price of one share, not the value of the entire company.

Enterprise value, or EV, represents the operating business for its capital providers. In our simplified examples, we move from EV to equity by subtracting financial debt and adding nonoperating cash and investments. **EBITDA** means earnings before interest, taxes, depreciation and amortization. It is an operating earnings measure, not cash available to shareholders.

2. Trailing and forward describe the period

Trailing uses earnings already reported. TTM and LTM mean trailing or last twelve months, usually the latest four reported quarters. FY means fiscal year. A fiscal-year result matches TTM only when their end dates align.

Forward uses forecast earnings. NTM means the next twelve months. FY1 usually refers to the next forecast fiscal year, but check the data provider’s convention. These labels matter because a forecast and an observed result carry different uncertainty.

My first check when comparing two P/Es is therefore simple: are both using the same earnings period? A trailing P/E for one company and a forward P/E for another are not a clean comparison.

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Apple: reading and using P/E

Let us separate two tasks. First, we can calculate the multiple embedded in an observed market price. Second, we can choose a benchmark multiple and use it to estimate a price. Those are different directions of the same calculation.

3. What multiple was the market paying?

Apple closed at **$270.37 on October 31, 2025**. Its reported FY2025 diluted EPS was **$7.46**. Divide $270.37 by $7.46 and you obtain **36.24× trailing P/E**. The market was paying about $36 for each dollar of that annual earnings figure.

Observed trailing P/E: $270.37 ÷ $7.46 = 36.24×

Now keep the stock price unchanged and assume that next-year EPS will be **$9.00**. The forward P/E becomes **30.04×**. The multiple falls because the denominator increases. Nothing happened to the stock price. The $9.00 forecast is my teaching assumption, not analyst consensus.

Assumed forward P/E: $270.37 ÷ $9.00 = 30.04×

The stock has not become cheaper: the denominator has increased. If earnings disappoint, the forecast-based multiple will have overstated that apparent advantage.

4. What price does a chosen multiple imply?

Suppose we decide that **30× trailing earnings** is an appropriate benchmark. Multiply Apple's $7.46 EPS by 30. The implied price is **$223.80**. Notice that I have matched a trailing benchmark with trailing earnings.

Implied price: 30 × $7.46 = $223.80 per share

Assumed trailing P/E	Actual FY2025 EPS	Implied price
25×	$7.46	$186.50
30×	$7.46	$223.80
35×	$7.46	$261.10

Choosing the multiple requires judgment. Stronger growth, sustainable profits or lower risk may justify a higher multiple. We must defend 30× rather than select it to obtain a desired answer.

Difference from market price: ($223.80 ÷ $270.37 − 1) × 100 = −17.23%

At $223.80, our scenario is about **17.2% below** the historical $270.37 market price. That difference is not a forecast of the return you will earn. It tells us what price follows from the assumed benchmark. We still need to defend that assumption.

Data: Apple FY2025 financial statements and Yahoo Finance AAPL historical prices. All dates are historical. The benchmark multiples and $9.00 forward EPS are illustrative assumptions.

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From enterprise value to equity

For EV/EBITDA and EV/Sales, we first value the operating business. We then identify the portion attributable to shareholders. I want you to keep that sequence clear so that you do not confuse enterprise value with equity value.

5. Identify the inputs and their units

Apple reported FY2025 operating income of **$133.050 billion** and depreciation and amortization of **$11.698 billion**. Adding them gives an operating EBITDA proxy of **$144.748 billion**. We assume a 25× EV/EBITDA benchmark for this example.

Every total amount below is in **billions of dollars**. Shares are in **billions of shares**. The debt is $7.979bn commercial paper plus $12.350bn current term debt plus $78.328bn long-term debt. Cash and investments comprise $35.934bn cash, $18.763bn current securities and $77.723bn noncurrent securities. These components explain the totals used in the bridge.

EBITDA proxy: 133.050 + 11.698 = $144.748bn

Financial debt: 7.979 + 12.350 + 78.328 = $98.657bn

Cash and securities: 35.934 + 18.763 + 77.723 = $132.420bn

6. Apply 25×, then identify the shareholder portion

Step	Calculation / result
1. Value operations	$144.748bn × 25 = **$3,618.700bn EV**
2. Deduct debt	Subtract **$98.657bn** of financial debt.
3. Add cash and investments	Add **$132.420bn**, giving **$3,652.463bn equity**.
4. Find the per-share value	Divide by 14.773260bn period-end shares: **$247.23 per share**.

Why subtract debt? EV values operations for capital providers; lenders have a claim that does not belong to common shareholders. **Why add nonoperating cash?** It is an asset in addition to operations. Apple has more cash and securities than financial debt in this simplified example, so equity ends up larger than EV.

Equity: 3,618.700 − 98.657 + 132.420 = $3,652.463bn

Per share: 3,652.463 ÷ 14.773260 = $247.23

The billions cancel when we divide dollars by shares. Do not divide by the share count twice, and do not subtract total liabilities instead of the specified financial debt. This example uses period-end shares to convert a total equity estimate into a per-share value.

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EV/Sales and the three answers

7. Start with the same company and a different denominator

Using Apple's **$416.161 billion** of FY2025 sales and an assumed **8×** multiple gives **$3,329.288 billion EV**. Subtract the same debt and add the same cash and investments. Equity becomes **$3,363.051 billion**. Divide by the same shares and the implied price is **$227.64**.

Step 1 · EV: 8 × 416.161 = $3,329.288bn

Step 2 · Equity: 3,329.288 − 98.657 + 132.420

= $3,363.051bn

Step 3 · Per share: 3,363.051 ÷ 14.773260 = $227.64

The assumed 8× means investors would pay $8 for every $1 of annual sales generated by the operating business. Because sales are measured before interest expense, we pair them with EV, which covers capital providers, rather than directly treating the result as shareholder value.

EV/Sales is easy to calculate, but sales alone tell us little about profitability. Two businesses can generate equal revenue while one earns strong margins and the other loses money. EV/EBITDA captures more of the cost structure, but still leaves out reinvestment and other cash obligations. This is why I would compare the result with our FCFE analysis.

8. Check what the three estimates actually mean

Method	Where the first calculation lands	Final price
30× P/E	Per-share equity value directly	$223.80
25× EV/EBITDA	Enterprise value, then equity	$247.23
8× EV/Sales	Enterprise value, then equity	$227.64

P/E uses earnings **after interest**, so the result already belongs to equity holders. We do not subtract debt again. The two EV methods use operating measures before interest, so we must make the bridge. That distinction is more important than memorizing the arithmetic.

For this classroom bridge, we treat all reported cash and marketable securities as nonoperating, use period-end shares, and omit leases and other claim adjustments. A full valuation would examine those choices. The inputs are from Apple's FY2025 statements; 25× and 8× are assumptions, not reported peer medians.

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Choosing a useful comparison

A peer is a company with sufficiently similar economics to make its multiple informative. Being in the same sector is only a starting point. Apple and Microsoft are both technology companies, but their products, revenue mix and margins differ. A device manufacturer may look closer to Apple in product terms but differ greatly in brand strength and service revenue.

9. How I would use a peer median

Imagine three suitable peers with trailing P/Es of **25×, 30× and 35×**. After sorting the observations, the middle value is **30×**: the median. Applying it to Apple's $7.46 trailing EPS gives the $223.80 estimate from our earlier example. These three peer multiples are hypothetical, so this illustrates the process rather than establishing Apple's fair value.

Sort the peers: 25×, 30×, 35×

Middle observation: 30×

Apply to Apple: 30 × $7.46 = $223.80

If a fourth comparable had a 60× P/E, the sorted set would be 25, 30, 35 and 60. With four observations, average the two middle numbers: the median becomes **32.5×**. The mean would be **37.5×**. Applied to the same EPS, those would imply $242.45 and $279.75 respectively. This shows why an outlier can change your estimate, even before you change the company's fundamentals.

A median is less sensitive to an extreme observation than a mean. But it does not rescue a poorly selected peer group. If a peer trades at a high multiple because its earnings are temporarily close to zero, I would investigate that denominator before treating the number as useful evidence.

10. Why our three answers differ

Method and assumption	Implied Apple price
P/E at 30×	$223.80
EV/EBITDA at 25×	$247.23
EV/Sales at 8×	$227.64

Each estimate uses a different measure of the business and a different assumed benchmark. I would not automatically average the answers. I would ask which peers and financial measures best reflect the business, whether earnings contain unusual items, and whether the implied growth and profitability are plausible.

For your comparison, align the observation date, trailing or forward period, accounting adjustments, currency and units. Negative earnings make P/E unhelpful as a conventional valuation benchmark. A low multiple can indicate weak prospects or high risk rather than a bargain.

The connection to DDM and FCFE is important: growth, profitability, reinvestment and risk are still driving value. Multiples put those expectations into a compact market benchmark instead of making every forecast explicit.

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Reading the trend: MA and RSI

We now change the question. Valuation asks what a business is worth. Technical analysis examines the history of market prices to describe trend and momentum. Traders may use this information when planning entries, exits and risk, but a technical signal does not calculate intrinsic value. Use the Apple charts in the lecture while reading this section.

11. Moving averages: a smoother view of price

The moving average smooths daily price fluctuations. In our chart, the 50-day average responds sooner than the 200-day average. When price is above a rising average, we have evidence of an upward trend over that horizon. We do not yet know whether the stock is undervalued.

On **December 31, 2025**, Apple closed at **$271.86**, compared with a 50-day average of **$272.66** and a 200-day average of **$231.75**. Price was slightly below its shorter-term average but well above its longer-term average. Both statements can be true because the horizons differ.

A crossover happens when one line moves through another. Traders may interpret a shorter average crossing above a longer one as strengthening trend. Remember that averages react to prices already observed. In a sideways market, repeated crossings can produce false starts and trading costs.

12. RSI: recent directional strength

RSI runs from 0 to 100. Our chart uses the conventional 14-day Wilder RSI. Readings above 70 are commonly called overbought, and readings below 30 oversold. Those words describe recent price momentum. They do not mean that we have measured fundamental overvaluation or undervaluation.

On **March 12, 2025**, Apple's RSI was approximately **29.9**. It was already below 30. Yet by April 8 its closing price had fallen another **20.5%** from the March 12 close. If you had treated "oversold" as an automatic buy instruction, you would have faced a substantial further decline.

My point is not that RSI never helps. It can draw attention to an unusually strong run of gains or losses. What it cannot do on its own is tell you that the move has ended. The example illustrates that limitation; it is not a backtest of a complete trading strategy.

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MACD and the full picture

MACD compares a faster price trend with a slower one. The lecture uses the standard 12/26/9 settings. You do not need to memorize the calculation here. Focus on what the lines and bars are telling you.

13. How I read the MACD chart

The blue line is MACD and the gold line is its signal line. When blue moves above gold, recent momentum is improving relative to the signal. When it moves below gold, momentum is weakening. The histogram bars show the gap between the lines. Smaller bars mean that gap is narrowing.

The zero line gives a separate comparison. On **June 30, 2025**, Apple's MACD was about **−0.31** while the signal line was about **−1.02**. MACD was above its signal, but still below zero. Momentum was improving even though the faster underlying trend remained below the slower one. There is no contradiction.

By **December 31**, MACD was about **−0.11**, below a signal line near **0.61**. RSI was about **45.6**, and price remained above the 200-day average. The longer-term trend reference and shorter-term momentum measures were describing different aspects of the same price history.

14. Putting the two parts of the lecture together

Start by deciding which question you are answering. To assess value, examine earnings, cash flows, reinvestment, risk and comparable firms. To describe market behavior, examine the price trend and momentum over a stated horizon. A stock can appear inexpensive on fundamentals while its price continues to fall.

Also remember that MA, RSI and MACD all use the same underlying price history. Agreement among them is not three independent pieces of evidence. Before relying on any trading rule, you would need to consider execution, costs, losses and performance on data not used to choose the rule.

There is no lecture activity to submit. The separate Canvas course feedback survey counts as Quiz 9. As you review, make sure you can explain why a multiple or an indicator gives the answer it does, rather than only repeating the number.

15. Sources and data conventions

Apple FY2025 financial statements; Yahoo Finance AAPL historical prices; Fidelity's guides to SMA, RSI and MACD. Charts use 2025 daily closing prices, with 2024 observations for indicator warm-up. Prices and price changes exclude dividends. Technical indicators are calculated from those closes. These are historical illustrations, not current recommendations.